

Sequences and series - Set 2 - Solutions

Specification MM2.1, MM2.2, MM2.3, M4.19.

Answer key

Question	Answer	Topic
1	A	Sequences and series
2	F	Sequences and series
3	A	Sequences and series
4	H	Sequences and series
5	B	Sequences and series
6	B	Sequences and series
7	C	Sequences and series
8	C	Sequences and series
9	A	Sequences and series
10	E	Sequences and series
11	H	Sequences and series

Worked solutions

Question 1 SEQ-16 · Sequences and series · spec MM2.2

The sum of the first n terms of an arithmetic series is S_n . Given that $S_{10} = 145$ and $S_{20} = 490$, find the common difference.

Answer: A

$$S_{10} = 5(2a + 9d) = 145 \Rightarrow 2a + 9d = 29; S_{20} = 10(2a + 19d) = 490 \Rightarrow 2a + 19d = 49. \text{ Subtract: } 10d = 20, d = 2.$$

Question 2 SEQ-18 · Sequences and series · spec MM2.1

A sequence is defined by $u_1 = 1$ and $u_{n+1} = u_n + 2n + 1$. Which of the following is an expression for u_n ?

Answer: F

$$u_2 = 4, u_3 = 9, u_4 = 16; u_n = n^2. \text{ Check: } (n+1)^2 = n^2 + 2n + 1.$$

Question 3 X2-04 · Sequences and series · spec MM2.3

A geometric series has first term $a \neq 0$ and common ratio r . P : ' $-1 < r < 1$ '. Q : 'the series has a sum to infinity'. Which one of the following is true?

Answer: A

A geometric series converges if and only if $|r| < 1$, which is exactly $-1 < r < 1$.

Question 4 X2-23 · Sequences and series · spec MM2.2

An arithmetic sequence has first term a and common difference d , both non-zero. Which of the following statements is/are true?

- I If $d > 0$ then every term after some point is positive.
- II The sum of the first n terms is a quadratic function of n .
- III If a and d are integers then S_n is an integer for every n .

Answer: H

I: terms $a + (n-1)d$ eventually exceed any bound. II: $S_n = \frac{n}{2}(2a + (n-1)d)$ is quadratic in n . III: S_n is a sum of integers.

Question 5 SEQ-05 · Sequences and series · spec MM2.3

A geometric series has first term a and common ratio r with $|r| < 1$. Its sum to infinity is 8 and the sum to infinity of the squares of its terms is $\frac{64}{3}$. Find r .

Answer: B

$$\frac{a}{1-r} = 8 \text{ and } \frac{a^2}{1-r^2} = \frac{64}{3}. \text{ Dividing the second by the square of the first: } \frac{1-r}{1+r} = \frac{64/3}{64} = \frac{1}{3}, \text{ so } 3-3r = 1+r \text{ and } r = \frac{1}{2}.$$

Question 6 SEQ-07 · Sequences and series · spec MM2.1

A sequence satisfies $x_{n+1} = \frac{1}{1-x_n}$ with $x_1 = 2$. Find x_{100} .

Answer: B

$$x_2 = \frac{1}{1-2} = -1, x_3 = \frac{1}{1-(-1)} = \frac{1}{2}, x_4 = \frac{1}{1-\frac{1}{2}} = 2 = x_1: \text{the sequence has period 3. Since } 100 = 3 \times 33 + 1, \\ x_{100} = x_1 = 2.$$

Question 7 SEQ-11 · Sequences and series · spec MM2.2

The sum of the first n natural numbers is S . The sum of the first $2n$ natural numbers is kS . Which of the following is true?

Answer: C

$$S = \frac{n(n+1)}{2} \text{ and the sum to } 2n \text{ is } \frac{2n(2n+1)}{2} = n(2n+1). \text{ So } k = \frac{n(2n+1)}{n(n+1)/2} = \frac{2(2n+1)}{n+1}.$$

Question 8 SEQ-17 · Sequences and series · spec MM2.3

A ball is dropped from a height of 10 m. After each bounce it rises to $\frac{3}{5}$ of its previous height. Find the total distance travelled by the ball before it comes to rest.

Answer: C

$$\text{Down: } 10 + 6 + 3.6 + \dots = \frac{10}{1-3/5} = 25. \text{ Up: } 6 + 3.6 + \dots = 15. \text{ Total } 40 \text{ m.}$$

Question 9 SEQ-19 · Sequences and series · spec MM2.3

The first three terms of a geometric sequence are x , $x+6$ and $x+15$. Find the fourth term.

Answer: A

$$\text{Geometric: } (x+6)^2 = x(x+15) \Rightarrow x^2 + 12x + 36 = x^2 + 15x \Rightarrow x = 12. \text{ The terms are } 12, 18, 27 \text{ with ratio } \frac{3}{2}, \text{ so the} \\ \text{fourth term is } 27 \times \frac{3}{2} = \frac{81}{2}.$$

Question 10 SEQ-20 · Sequences and series · spec MM2.2

The integers from 1 to 100 inclusive are written down. What is the sum of those that are multiples of 3 or 5 (or both)?

Answer: E

$$\text{Multiples of 3: } 3 + \dots + 99 = 3(1 + \dots + 33) = 3 \times 561 = 1683. \text{ Multiples of 5: } 5(1 + \dots + 20) = 1050. \text{ Multiples} \\ \text{of 15: } 15(1 + \dots + 6) = 315. \text{ Total } 1683 + 1050 - 315 = 2418.$$

Question 11 SEQ-21 · Sequences and series · spec MM2.1

A sequence has $u_1 = 3$ and $u_{n+1} = \frac{u_n + 5}{2}$. Which of the following statements is/are true?

I The sequence is increasing.

II $u_n < 5$ for all n .

III $u_n = 5 - 2^{2-n}$ for all n .

Answer: H

$$u_2 = 4, u_3 = 4.5, u_4 = 4.75. \text{ If } u_n < 5 \text{ then } u_{n+1} = \frac{u_n + 5}{2} < 5 \text{ and } u_{n+1} - u_n = \frac{5 - u_n}{2} > 0, \text{ so I and II hold. III: } 5 - 2^{2-n} \\ \text{gives } 3, 4, 4.5, \dots \text{ and satisfies the recurrence: } \frac{5 - 2^{2-n} + 5}{2} = 5 - 2^{1-n}. \text{ All three are true.}$$